

# Cities

Week 6

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## Today

Two papers run the same experiment in opposite directions.

1. **Three theories** of why economic activity sits where it sits. They disagree about what a shock would do.
2. **Destroy the city, keep the site.** Davis and Weinstein use the Allied bombing of Japan.
3. **Keep the city, delete the site.** Bleakley and Lin use the railroad, which made a set of river towns pointless.

Why do cities exist at all? What fixes the spatial distribution of economic activity? And why is that distribution so extraordinarily persistent?

**Three theories of where cities are**

## Why is there a city there?

Why Pittsburgh, why Dubai, and why is Minneapolis larger than St. Paul?  
Economists give three answers.

| Theory                  | Why the city is there                                       | What would move it    |
|-------------------------|-------------------------------------------------------------|-----------------------|
| Locational fundamentals | the site is genuinely best<br>harbor, river, coal, soil     | changing the site     |
| Increasing returns      | everyone else is already there<br>spillovers, labor pooling | a big enough shock    |
| Random growth           | no reason in particular<br>cities as a random walk          | anything, permanently |

Taxonomy after Christina and David Romer's urban economic history lectures.

## Locational fundamentals

### The claim

Some sites are more productive than others because of fixed features of geography, such as a harbor, navigable water, coal, fertile soil or a mild climate. People and firms go to the best sites, so a city's size tracks the value of its site.

- **New York.** A deep, sheltered harbor at the mouth of the Hudson, and the Hudson–Mohawk corridor, the one water-level route through the Appalachians, which the Erie Canal followed to the Great Lakes in 1825.
- **Pittsburgh.** The point where the Allegheny and the Monongahela form the Ohio, on top of the Pittsburgh coal seam.
- **Chicago.** The short portage from the Chicago River to the Des Plaines, a headwater of the Illinois, which linked the Great Lakes to the Mississippi system.

A shock to a city's people leaves its site intact, so the city comes back. A change in the value of the site moves the city.

Chicago is a portage site of the kind Bleakley and Lin study when they extend their fall-line design to the Midwest.

## The United States at night, 2012



Find the coasts, the shore of the Great Lakes and the river valleys of the interior. NASA Earth Observatory image by Robert Simmon, using Suomi NPP VIIRS data provided courtesy of Chris Elvidge (NOAA National Geophysical Data Center). Public domain.

## How much geography explains

- **Rappaport and Sachs (2003).** Counties whose centers lie within 80 km of an ocean or Great Lakes coast cover 13 percent of the continental United States. In 2000 they held 51 percent of its population and 57 percent of its income.
- **Henderson, Squires, Storeygard and Weil (2018).** Twenty-four physical-geography variables explain 47 percent of the variation in nightlights across 240,000 grid cells worldwide and 35 percent of the variation within countries.
- **Ellison and Glaeser (1999).** A few observable natural advantages explain about 20 percent of the geographic concentration of American manufacturing industries. They conjecture that better measures would raise the share to at least half.

Geography variables account for a large share of where activity sits. That share also counts places geography once favored and history has kept, so it is an upper bound on what today's fundamentals explain.

Henderson and coauthors also find that within countries that developed early, agricultural variables explain more than six times as much of the variation in lights as trade variables. Among late developers the ratio is about 1.5.

## Why proximity raises productivity

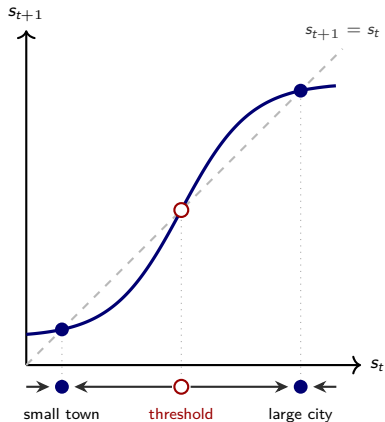
Marshall's account of localized industry is usually distilled into three sources of advantage.

- **Input sharing.** “Subsidiary trades grow up in the neighbourhood, supplying it with implements and materials.” A specialized supplier needs a large local market.
- **Labor pooling.** A localized industry “offers a constant market for skill.” Workers and employers match faster where there are many of both.
- **Knowledge spillovers.** “The mysteries of the trade become no mysteries; but are as it were in the air.”

Each source makes a place more productive the more activity it already has. Where a city sits then depends on its history as well as on its site.

Marshall, *Principles of Economics*, 8th edition (1920), Book IV, chapter X. Duranton and Puga (2004) restate the three as sharing, matching and learning. Krugman's *Geography and Trade* (1991) traces the carpet industry of Dalton, Georgia, to a tufted bedspread a teenager named Catherine Evans made as a wedding gift in 1895.

## Increasing returns allow more than one equilibrium



- The curve maps a city's share of activity today,  $s_t$ , into its share next period. Where it crosses the 45-degree line the share stays put.
- Agglomeration makes the curve steeper than the 45-degree line in the middle, so the middle crossing repels. Above the threshold a city gains share until it is the large city, and below it a city loses share until it is the small town.
- A small shock to the large city wears off. A shock that pushes it below the threshold sends it to the small equilibrium, and it stays there.

The diagram is stylized. Filled dots are stable equilibria and the hollow dot is the unstable threshold. In Krugman (1991) whether a country splits into an industrial core and an agricultural periphery depends on transport costs, scale economies and manufacturing's share of income. He offers the model as a rigorous version of the circular story geographers told about the American manufacturing belt.

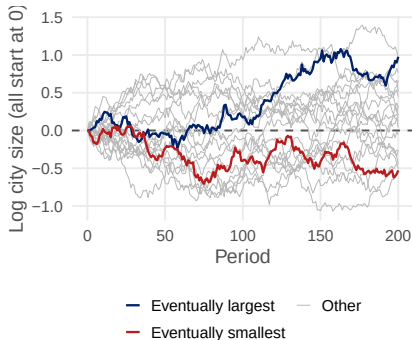
## Evidence that proximity pays

- **Greenstone, Hornbeck and Moretti (2010).** They compare counties that won a large new manufacturing plant with the runner-up counties the firm nearly chose. Productivity trends were similar before the opening. Five years after it, incumbent plants' TFP is 12 percent higher in the winning counties, and the gain is larger for plants that share labor and technology with the new one.
- **Ciccone and Hall (1996).** Across American states a doubling of employment density raises average labor productivity by about 6 percent.
- **Hornbeck and Keniston (2017).** The Boston Fire of 1872 let owners rebuild together, and upgrades on one plot encouraged upgrades nearby. Land values rose substantially on burned plots and on unburned plots near them.

These estimates establish that proximity pays. Whether a shock can tip a city into a new equilibrium is a separate question, and Davis and Weinstein and then Bleakley and Lin take it up.

The runner-up county is the counterfactual, which is the Week 2 design applied to plant location.

## Gibrat's law

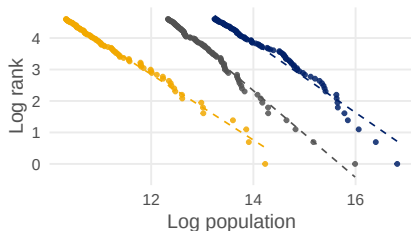


- Every city draws its growth rate from the same distribution whatever its size. Log size is then a random walk.
- A random walk has no level to return to, so every shock is permanent. With no floor on size the spread of sizes widens forever.
- The theory needs no reason for any particular place. The twenty cities in the figure start equal and end far apart on luck alone.

Simulated. Twenty cities, 200 periods, log growth drawn from a normal distribution with standard deviation 0.05.

Code in `scripts/R/wk06_city_size.R`.

## Zipf's law for American cities



- Cities, 1890 (slope  $-1.04$ )
- Metro areas, 2020 (slope  $-1.12$ )
- City limits, 2020 (slope  $-1.38$ )

- Rank a country's cities by population. Zipf's law says rank times size is roughly constant, so the second city is about half the first and a regression of log rank on log size has a slope near  $-1$ .
- The 100 largest cities of 1890 give  $-1.04$  and the 100 largest metro areas of 2020 give  $-1.12$ . With standard errors near  $0.15$  neither is distinguishable from  $-1$ .
- City limits in 2020 give  $-1.38$ , about two standard errors from  $-1$ . Boundaries split each metro area into many places, so the largest cities lose their suburbs and the suburbs enter the list as cities.

Slopes from regressing log rank on log population, standard errors  $|\text{slope}| \sqrt{2/n}$  (Gabaix and Ibragimov 2011).

Their small-sample correction,  $\log(\text{rank} - 1/2)$ , makes each slope about  $0.07$  steeper. Census Working Paper 27 (Gibson 1998), Table 12, for 1890, when Brooklyn was still separate from New York. Census estimates base of 1 April 2020 for metro areas and places. Code in `scripts/R/wk06_city_size.R`.

## What random growth explains

- **Gabaix (1999).** If cities grow by the same proportional process whatever their size and a floor stops small cities from vanishing, the distribution converges to a power law. Its exponent approaches 1, Zipf's value, as the floor shrinks relative to mean size.
- **Eeckhout (2004).** Across all 25,359 American places in 2000 the size distribution is lognormal. Zipf's law fits its upper tail, and growth is independent of size. Levy (2009) finds that the top 0.6 percent of places follow a power law instead.

Random growth explains the shape of the size distribution and is silent on which places are large. Tonight's two papers ask which places are large and why they stay large.

## Power laws and a second route to Zipf's law

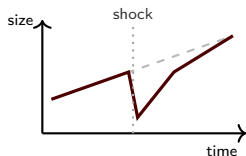
### Power law

A size distribution follows a power law when the number of cases larger than  $S$  is proportional to  $S^{-\zeta}$ . Raising the threshold by a given factor always cuts the count by the same fraction, so there is no typical size, and in logs the relation is a straight line with slope  $-\zeta$ . Zipf's law is  $\zeta = 1$ . There are then about twice as many cities above 500,000 as above one million and ten times as many above 100,000.

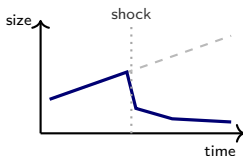
- **Random growth puts the power law in the cities.** Population itself grows by proportional shocks, and the distribution settles at Zipf's law.
- **Krugman (1996) puts it in the sites.** Suppose the value of sites (harbors, hinterlands, the area a river drains) follows a power law and each city's size tracks the value of its site. The cities then inherit the law with no randomness in their own growth.
- **Davis and Weinstein count this in favor of fundamentals.** On that assumption fundamentals explain the shape of the size distribution and why particular places come back after a shock. Random growth explains only the first, and increasing returns has no simple account of the first.

Davis and Weinstein posit the distribution of site values and do not test it (pp. 1270–1271). Bleakley and Lin's portage sites are a concrete case, since a site's value rises with the size of the watershed above it.

## The theories disagree about what a shock does

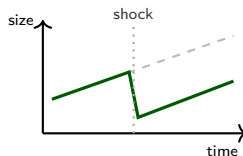


fundamentals, or increasing  
returns after a small shock



**falls to a new equilibrium**

increasing returns after  
a shock past the threshold



**stays where it landed**

random growth

- **Locational fundamentals.** Knock the people out and the city comes back, because the site is still the best site.
- **Increasing returns.** A small shock wears off. A shock past the threshold sends the city to a different equilibrium, and it stays there.
- **Random growth.** Growth is independent of size, so no shock ever wears off.

History ran this experiment twice, once in the bombing of Japan and once when railroads made portage obsolete.

**Eight thousand years of Japanese settlement**

## Regional population from archaeological site counts

- Regional population for **39 Japanese regions**, from the Stone Age to 1998. The early numbers come from archaeological site counts, which is where the bones in the title come from.
- Two findings. There is a great deal of variation in regional density, and it is old. And the identity of the dense regions is remarkably stable — rank correlations with 1998 stay high across enormous spans of time.
- Regions that were densely settled before writing arrived in Japan are disproportionately the densely settled regions now.

Persistence over eight millennia is the case for taking locational fundamentals seriously. Harbors, plains and rivers do not move.

Davis and Weinstein (2002), “Bones, Bombs, and Break Points.”

## Regional population in Japan, 6000 BC to 1998

| Year          | Population<br>(thousands) | Share of<br>top five | Relative<br>variance | Zipf<br>(s.e.) | Raw corr.<br>with 1998 | Rank corr.<br>with 1998 |
|---------------|---------------------------|----------------------|----------------------|----------------|------------------------|-------------------------|
| 6000–300 BC   | 125                       | 0.39                 | 2.46                 | −0.809 (0.217) | 0.53                   | <b>0.31</b>             |
| 300 BC–AD 300 | 595                       | 0.23                 | 0.93                 | −1.028 (0.134) | 0.67                   | 0.50                    |
| 725           | 4,511                     | 0.20                 | 0.72                 | −1.207 (0.133) | 0.60                   | 0.71                    |
| 800           | 5,506                     | 0.18                 | 0.75                 | −1.184 (0.152) | 0.57                   | 0.68                    |
| 900           | 7,442                     | 0.29                 | 0.68                 | −1.230 (0.166) | 0.48                   | 0.65                    |
| 1150          | 6,836                     | 0.20                 | 0.66                 | −1.169 (0.141) | 0.53                   | 0.73                    |
| 1600          | 12,266                    | 0.30                 | 0.64                 | −1.192 (0.068) | 0.76                   | 0.83                    |
| 1721          | 31,290                    | 0.21                 | 0.43                 | −1.582 (0.113) | 0.85                   | 0.84                    |
| 1798          | 30,531                    | 0.21                 | 0.37                 | −1.697 (0.120) | 0.83                   | 0.81                    |
| 1872          | 33,748                    | 0.18                 | 0.30                 | −1.877 (0.140) | 0.76                   | 0.78                    |
| 1920          | 53,032                    | 0.25                 | 0.43                 | −1.476 (0.043) | 0.94                   | 0.93                    |
| 1998          | 119,486                   | 0.41                 | 1.00                 | −0.963 (0.025) | 1.00                   | 1.00                    |

Davis and Weinstein (2002), Table 1, 39 regions. The rank-correlation column is the one to read. It is 0.31 for regions dense before 300 BC and never below 0.65 from the eighth century onward. Relative variance is the variance of log density in each year divided by its 1998 value. Japan was closed to most foreign trade from the 1630s to the 1850s, so the 1721 and 1798 rows fall inside that period and 1872 comes just after it. Regional density grew more even as port regions lost population, which shows in the low relative variance and steep Zipf coefficients of those rows.

## Persistence cannot separate the theories

- Persistence is consistent with **fundamentals**. The site was good and still is.
- Persistence is equally consistent with **increasing returns**. People are there because people were there, and the original reason stopped mattering centuries ago.
- In peacetime the two are observationally identical. You cannot separate them by looking at how correlated a place is with its own past.

Bleakley and Lin call this **state dependence** versus **serial correlation**. The things that first attracted people are usually still there, which confounds any attempt to attribute today's map to history rather than to geography.

Separating them takes a shock, ideally two pointing opposite ways.

## State dependence and serial correlation

- **State dependence.** Past population causes today's population. People are there because people were there, which is the increasing-returns story.

$$P_{it} = \rho P_{i,t-1} + \varepsilon_{it}$$

- **Serial correlation.** A persistent advantage of the site,  $\mu_i$ , causes population in every period, and past population has no effect of its own. This is the locational-fundamentals story.

$$P_{it} = \mu_i + \varepsilon_{it}$$

- Both models make population today highly correlated with population a century ago, so persistence alone cannot tell them apart. Random growth is the extreme case of state dependence, with  $\rho = 1$  and no  $\mu_i$ .

Each paper removes one of the two causes. Davis and Weinstein remove the population and keep  $\mu_i$ , and the cities came back, which points to serial correlation. Bleakley and Lin remove  $\mu_i$  and keep the population, and the towns stayed, which points to state dependence.

$P_{it}$  is the population of place  $i$  in period  $t$ . Bleakley and Lin define state dependence as “the presence of factors of production” and serial correlation as “the advantages that first attracted other factors.”

**The bombing of Japan as a temporary shock**

## The first experiment

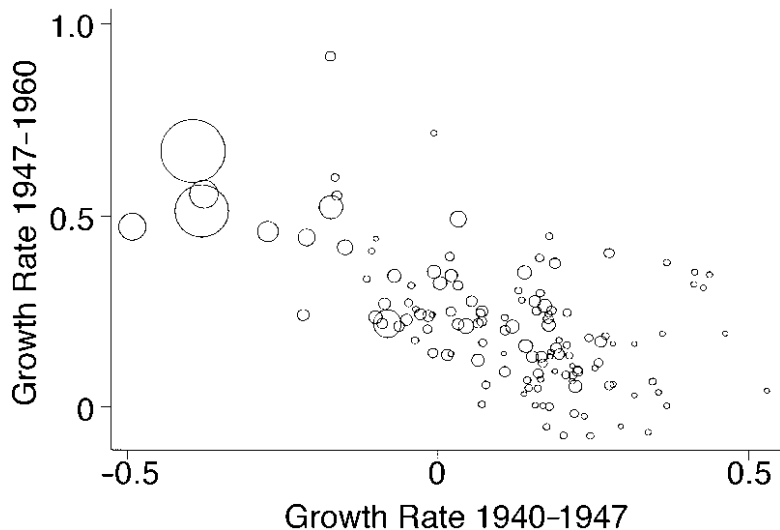
- Allied bombing devastated **66** Japanese cities. It destroyed almost half of all structures in them — **2.2 million buildings**. About **300,000** people were killed and **40 percent** of the population of those cities was made homeless. Some cities lost as much as half their people to death, flight and disappearance.
- The single worst night was the firebombing of Tokyo on 9 March 1945, when more than three hundred B-29s flew low with maximum loads of napalm incendiaries over a city that had seen no rain in a week.

## Why it is a usable experiment

- The fundamentals were untouched. The harbors, the rivers, the plains and the coal were all exactly where they had been in 1940.
- The accumulated factors were destroyed. That means people, housing, factories, rail yards and the whole physical stock that increasing-returns theories say holds a city in place.
- Severity varied a great deal across cities, often for reasons unrelated to their prospects. Kyoto was spared for its history, Niigata and Kitakyushu were held back as atomic targets, and the north lay out of bomber range.

The design takes the increasing returns away, leaves the fundamentals alone and asks whether the city comes back.

## Cities that lost the most grew back the fastest



Davis and Weinstein (2002), Figure 1. Growth 1947–1960 against growth 1940–1947 for cities above 30,000 people with positive casualties, circle size proportional to 1925 population. The negative slope is the recovery.

## The Davis and Weinstein model

Let  $s_{it}$  be city  $i$ 's share of national population.

$$s_{it} = \mu_i + \varepsilon_{it} \qquad \varepsilon_{i,t+1} = \rho \varepsilon_{it} + \nu_{i,t+1}$$

$\mu_i$  is the **permanent** component, the site.  $\varepsilon_{it}$  is everything else, and a fraction  $\rho$  of it carries into the next period.

$$\begin{aligned} s_{i,t+1} - s_{it} &= \varepsilon_{i,t+1} - \varepsilon_{it} && \text{difference, } \mu_i \text{ cancels} \\ &= (\rho - 1) \varepsilon_{it} + \nu_{i,t+1} && \text{substitute } \varepsilon_{i,t+1} \\ &= (\rho - 1) \nu_{it} + [\nu_{i,t+1} + \rho(\rho - 1) \varepsilon_{i,t-1}] && \text{substitute } \varepsilon_{it} \end{aligned}$$

Set  $t = 1947$ . Then  $\nu_{i,1947}$  is the wartime shock, the left side is growth from 1947 to 1960, and the bracket is the error term.

Regressing postwar growth on the wartime shock estimates  $\rho - 1$  directly.

|                                     | $\rho$ | coefficient |
|-------------------------------------|--------|-------------|
| Random growth, shocks are permanent | 1      | 0           |
| Shocks fully dissipate              | 0      | -1          |

## Why it has to be instrumented

- The regressor we want is the shock. The regressor we have is growth from 1940 to 1947, which is the shock plus whatever the city's own trend was doing.
- The regressor therefore carries measurement error and an omitted trend at once, and both correlate it with the error term.
- Davis and Weinstein instrument the wartime growth rate with two measures of the bombing, **deaths per capita** and **buildings destroyed per capita**, both scaled by 1940 population.

That is the third instrumental variable of the term, after settler mortality in Week 3 and the speed of westward expansion in Week 4. Ask the Week 2 questions of it. Is it relevant, and is the exclusion restriction plausible?

# The first stage

TABLE 2—INSTRUMENTAL VARIABLES EQUATION  
(DEPENDENT VARIABLE = RATE OF GROWTH IN CITY  
POPULATION BETWEEN 1940 AND 1947)

| Independent variable           | Coefficient       |
|--------------------------------|-------------------|
| Constant                       | 0.213<br>(0.006)  |
| Deaths per capita              | -0.665<br>(0.506) |
| Buildings destroyed per capita | -2.335<br>(0.184) |
| $R^2$ :                        | 0.409             |
| Number of observations:        | 303               |

*Note:* Standard errors are in parentheses.

Davis and Weinstein (2002), Table 2. 303 cities.

- Together the two instruments explain **41 percent** of the variance in wartime city growth. Relevance is secure.
- Buildings destroyed per capita does the work, at  $-2.335$  with a standard error of  $0.184$ . Deaths per capita is  $-0.665$  with a standard error of  $0.506$ , so on its own it is not distinguishable from zero.
- Destroying the housing stock moved people. Deaths, dreadful as the numbers are, moved the population share less.

# The two-stage least-squares estimates

TABLE 3—TWO-STAGE LEAST-SQUARES ESTIMATES OF  
IMPACT OF BOMBING ON CITIES  
(INSTRUMENTS: DEATHS PER CAPITA AND BUILDINGS  
DESTROYED PER CAPITA)

| Independent variable                               | Dependent variable =<br>growth rate<br>of population<br>between<br>1947 and<br>1960 |                   | Dependent variable =<br>growth rate<br>of population<br>between<br>1947 and<br>1965 |
|----------------------------------------------------|-------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------------------------------|
|                                                    | (i)                                                                                 | (ii)              | (iii)                                                                               |
| Growth rate of population<br>between 1940 and 1947 | −1.048<br>(0.097)                                                                   | −0.759<br>(0.094) | −1.027<br>(0.163)                                                                   |
| Government reconstruction<br>expenses              | 1.024<br>(0.387)                                                                    | 0.628<br>(0.298)  | 0.392<br>(0.514)                                                                    |
| Growth rate of population<br>between 1925 and 1940 |                                                                                     | 0.444<br>(0.054)  | 0.617<br>(0.092)                                                                    |
| $R^2$ :                                            | 0.279                                                                               | 0.566             | 0.386                                                                               |
| Number of observations:                            | 303                                                                                 | 303               | 303                                                                                 |

- **Column (i).** The coefficient is  $-1.048$  with a standard error of  $0.097$ , so  $\rho$  is about zero and the typical city had recovered its relative size within fifteen years.
- **Column (ii)** adds growth from 1925 to 1940 to control for the possibility that the United States bombed fast-growing cities. The coefficient falls to  $-0.759$ , so three quarters of the loss was recovered by 1960.
- **Column (iii)** keeps that control, extends the window to 1965 and gets  $-1.027$ . Given the prewar trend, the damage is entirely reversed.

Davis and Weinstein (2002), Table 3. Reconstruction spending is positive and small, and significant in columns (i) and (ii). A one-standard-deviation increase raises 1960 city size by about 2.2 percent.

## Japan's cities returned to their prewar ranks

- Most cities were back in their prewar position in the size distribution within about fifteen years.
- With the window extended to 1965 the coefficient reaches  $-1.027$ . In their words, “the effect of the temporary shocks vanishes completely in less than 20 years.”
- The refugee objection fails too. Recovery driven only by displaced people going home would be weakest in Hiroshima and Nagasaki, and they recovered as well.

The bombing was about as large a temporary shock as any set of cities has taken, and it left no lasting mark on relative city size. Increasing returns allowed that a hard enough push could tip a city into a new equilibrium. Japan's cities were pushed about as hard as cities are pushed, and on average they came all the way back.

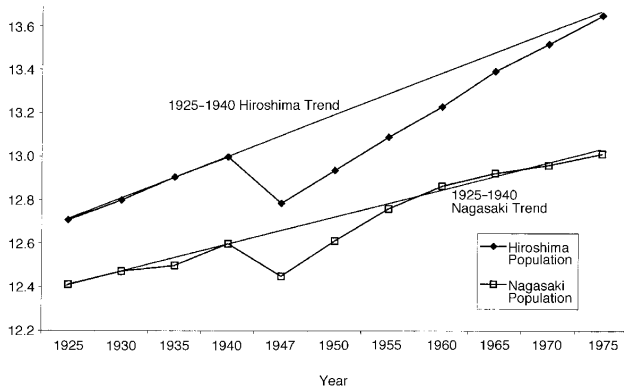
## Hiroshima and Nagasaki

|                                  | Hiroshima           | Nagasaki           |
|----------------------------------|---------------------|--------------------|
| Date                             | 6 August 1945       | 9 August 1945      |
| Yield                            | 12 kilotons         | 22 kilotons        |
| Built-up area destroyed          | over two thirds     | under 40 percent   |
| Killed                           | as many as 80,000   | more than 25,000   |
| Share of population              | <b>20.8 percent</b> | <b>8.5 percent</b> |
| Rank among Japanese cities, 1940 | 8th                 | 12th               |

- Nagasaki took the larger weapon and the smaller loss, because the ground is hilly and the bomb missed its aiming point by a wide margin.
- They also test an objection. Perhaps recovery was only displaced people drifting home along social ties, which would be weakest where the most people died.

Across the 144 cities with positive casualties the average death rate was about 1 percent.

## Even these two returned to trend



Davis and Weinstein (2002), Figure 2. Hiroshima and Nagasaki against their own 1925–1940 trend lines. Both fall off a cliff and both climb back onto the line. Hiroshima, which lost the larger share of its people, took somewhat longer.

**The railroad as a permanent shock**

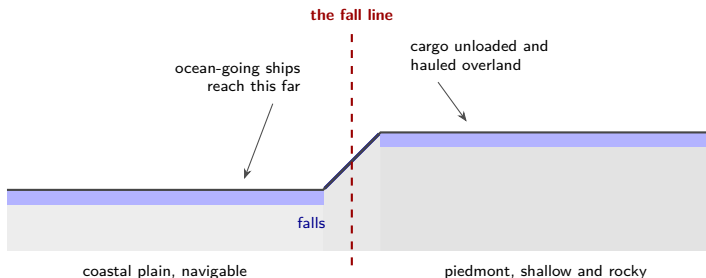
## The mirror-image experiment

- The bombing destroyed the factors and left the fundamental in place. The city returned, which points at fundamentals.
- To finish the identification you want the mirror image. Leave the city standing and take the fundamental away. If the city persists anyway, the site was never what was holding it there.
- History supplies that experiment too. The railroad took the fundamental away.

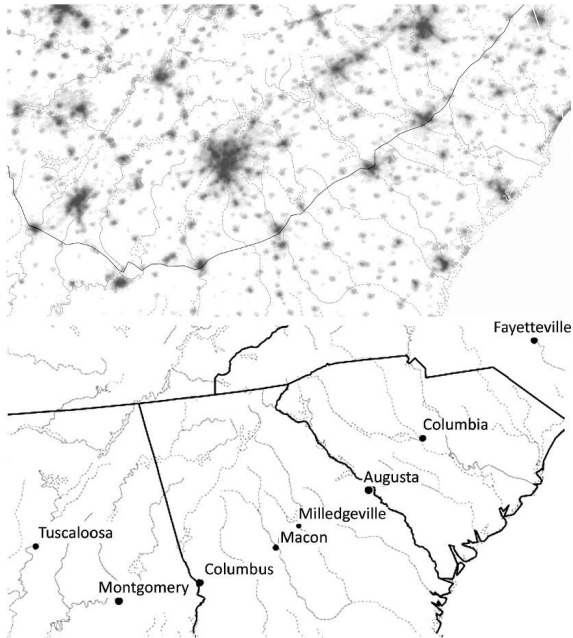
Neither paper on its own can separate the theories. Together they can.

Bleakley and Lin (2012), "Portage and Path Dependence."

## What a portage site is



- The **fall line** is where the hard rock of the piedmont meets the soft coastal plain, and rivers cross it in a final set of rapids. It runs from New Jersey to Alabama and on west into Texas.
- Ships came up from the sea and stopped. Cargo was unloaded and hauled — **portaged** — past the falls. So warehouses, agents, insurers, teamsters and, where the falls gave water power, mills all accumulated at that one point.



Bleakley and Lin (2012), Figure II. Nightlights in 2003 above, and the fall line and its cities below. They still sit in a line along a feature that stopped mattering a century ago.

# The portage advantage disappeared between 1850 and 1930

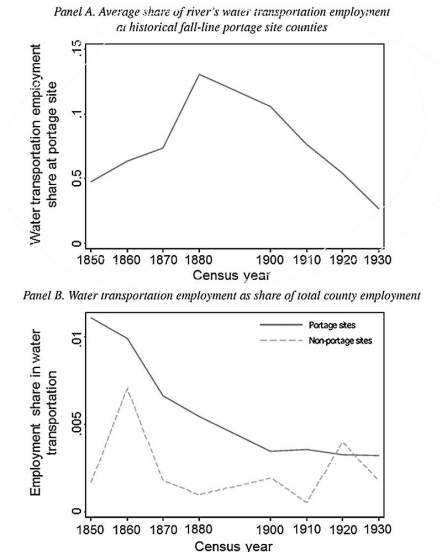


FIGURE I

Bleakley and Lin (2012), Figure I. Water-transportation employment across 51 historical portage sites. Railroads, and then trucks, made the portage worthless. By 1930 the original advantage is gone.

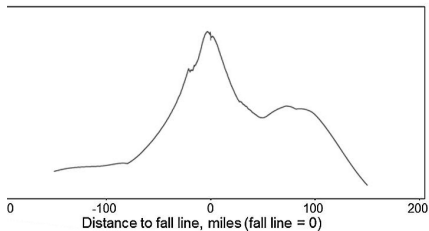
## On fundamentals alone the towns should have gone too

- The fundamental that created these places was a **transshipment bottleneck**, and the bottleneck was abolished outright.
- Water power went the same way, first to steam and then to electricity transmitted from anywhere.
- If cities sit where they sit because the site is best, a site that stops being best should stop having a city on it.

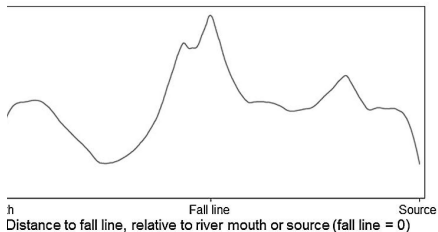
The obsolescence is total and dated. Portaging had stopped mattering by 1930, nearly a century ago.

# Population density peaks at the fall line

Panel A: Average by absolute distance from the fall line



Panel B: Average by renormalized distance from the fall line



Bleakley and Lin (2012), Figure III. Population density in 2000 along fall-line rivers, against distance from the fall line. The spike sits at zero. Panel B renormalizes distance by position between river mouth and source, in case proximity to the coast is doing the work, and the spike survives.

## The specification

$$\ln \text{density}_{gr} = \beta \text{portage}_g + \alpha_1 D_g^{FL} + \alpha_2 D_g^R + Z_g \xi + \delta_r + \epsilon_{gr}$$

- $g$  is a small area — a census tract, a county or a pixel of nightlights — sitting in river watershed  $r$ .
- **portage** is a dummy for the area's centroid being within fifteen miles of a river crossing the fall line.
- $D^{FL}$  is a dummy for being near the fall line.  $D^R$  is a dummy for being near a river.
- $\delta_r$  is a **watershed fixed effect**, and  $Z_g$  includes a fourth-order polynomial in latitude and longitude. Standard errors are clustered on the 53 watersheds.

A portage site is where the fall line crosses a river, so the portage dummy is an **interaction**, and both of its first-order terms are in the regression.

So  $\beta$  measures what having both features adds beyond each one alone. That is a difference in differences laid out across a map instead of across time.

# The results

TABLE I  
PROXIMITY TO HISTORICAL PORTAGE SITE AND CONTEMPORARY POPULATION DENSITY

|                                                      | (1)<br>Basic         | (2)<br>Other spatial controls | (3)                            | (4)                  | (5)<br>Additional fixed factors | (6)                  | (7)                  | (8)                           |
|------------------------------------------------------|----------------------|-------------------------------|--------------------------------|----------------------|---------------------------------|----------------------|----------------------|-------------------------------|
|                                                      |                      | State fixed effects           | Distance from various features | Climate variables    | Aquifer Share                   | Mean elevation       | Atlantic Rivers only | Within 100mi of the fall line |
| Specifications:                                      |                      |                               |                                |                      |                                 |                      |                      |                               |
| Explanatory variables:                               |                      |                               |                                |                      |                                 |                      |                      |                               |
| <i>Panel A: Census Tracts, 2000, N = 21452</i>       |                      |                               |                                |                      |                                 |                      |                      |                               |
| Dummy for proximity to portage site                  | 1.113<br>(0.340)***  | 1.009<br>(0.321)***           | 1.118<br>(0.243)***            | 1.041<br>(0.316)***  | 0.979<br>(0.330)***             | 1.077<br>(0.316)***  | 0.838<br>(0.401)**   | 1.039<br>(0.319)***           |
| Distance to portage site, natural logs               | -0.617<br>(0.134)*** | -0.653<br>(0.128)***          | -0.721<br>(0.118)***           | -0.460<br>(0.121)*** | -0.562<br>(0.123)***            | -0.577<br>(0.118)*** | -0.572<br>(0.177)*** | -0.764<br>(0.142)***          |
| <i>Panel B: Nighttime Lights, 1996-97, N = 65000</i> |                      |                               |                                |                      |                                 |                      |                      |                               |
| Dummy for proximity to portage site                  | 0.504<br>(0.144)***  | 0.445<br>(0.127)***           | 0.490<br>(0.161)***            | 0.500<br>(0.144)***  | 0.506<br>(0.147)***             | 0.522<br>(0.155)***  | 0.495<br>(0.151)***  | 0.391<br>(0.100)***           |
| Distance to portage site, natural logs               | -0.188<br>(0.065)*** | -0.159<br>(0.065)**           | -0.151<br>(0.090)              | -0.186<br>(0.061)*** | -0.196<br>(0.065)***            | -0.138<br>(0.059)**  | -0.130<br>(0.101)    | -0.212<br>(0.060)***          |

- Being within 15 miles of a portage site is worth **1.1 log points** of census-tract density. Since  $e^{1.113} \approx 3.0$ , these tracts are about three times as dense as comparable tracts in the same watershed, with a 95 percent interval of 1.6 to 5.9 times. That is a large gap for an advantage that disappeared about a century ago. In nightlights the coefficient is 0.5, or about 66 percent brighter.
- The other columns add state fixed effects, distances to the fall line, the ocean and the nearest 1890 seaport, and controls for climate, aquifer share and elevation. Others restrict the sample to Atlantic rivers or to within 100 miles of the fall line. Across all eight the estimate stays between 0.84 and 1.12 and remains significant.
- Distance to the portage in logs is negative throughout, which is the same fact read as a gradient rather than a jump.

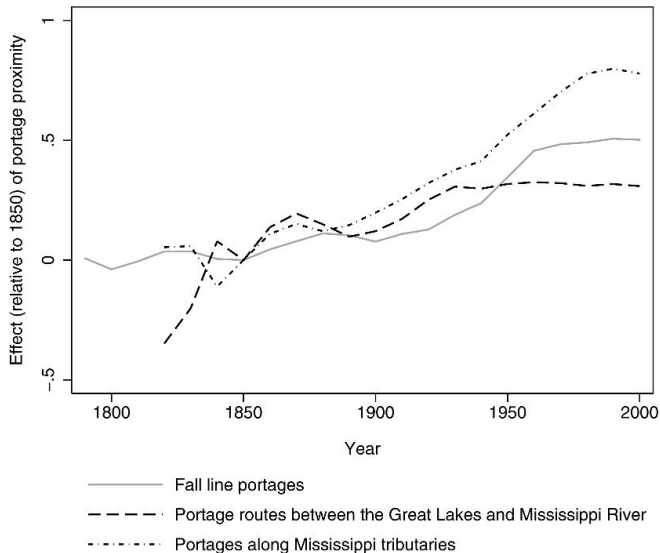
Bleakley and Lin (2012), Table I. Panel A is 21,452 census tracts, Panel B is 65,000 nightlight observations. Their text (p. 607) reports the dummy as predicting “50% to 110% increases,” which reads log points as percentages. A coefficient of 1.1 is an increase of about 200 percent.

## Alternative explanations

- **Maybe fall-line sites are just good sites.** Bleakley and Lin check the observables and find no advantage. Richmond, as they put it, is essentially as flat as the land nearby, so a road-building story does not work.
- **Maybe it is water power.** That mattered in 1820 and stopped mattering with steam and the electrical grid.
- **Maybe it is just the river.** Figure III compares points along the same rivers. Every point on a river has the river. Only the fall line had the portage.

The fourth explanation is **sunk costs**. Perhaps the buildings, wharves and streets are durable capital that is still slowly depreciating. That would be persistence without any increasing returns at all.

## The portage effect has grown since 1850



Bleakley and Lin (2012), Figure V. The effect of portage proximity on population density by decade, 1790–2000, measured relative to 1850. The elasticity is larger today than when portaging still had direct value.

## Sunk costs against increasing returns

Sunk capital depreciates. Wharves rot, warehouses are demolished, streets are rebuilt somewhere else. A sunk-cost explanation predicts the portage advantage should fade.

Instead it is stronger in 2000 than in 1850, when the portage was still doing real work. Something is adding to the advantage over time, and that something is other people and other firms.

- A permanent shock to fundamentals did not move these cities either.
- Once the pile of activity exists, the pile is its own reason. That is increasing returns.

**Putting the two together**

## Two experiments with apparently opposite answers

|          | Davis and Weinstein | Bleakley and Lin          |
|----------|---------------------|---------------------------|
| Shock    | bombing             | obsolescence of portage   |
| Hits     | people, buildings   | the site's advantage      |
| Nature   | temporary           | permanent                 |
| Result   | full recovery       | no decay at all           |
| Reads as | fundamentals matter | increasing returns matter |

- Both cannot be the whole story, so what reconciles them?
- **Heterogeneity of the fundamentals.** Japan is mountainous, narrow and harbor-scarce. Its good sites are obviously better, and they pin the map down. The American fall-line region is comparatively uniform. When no site is much better than its neighbor, history and accumulation decide which one grows.

Davis and Weinstein's own conclusion is already a hybrid. Fundamentals establish the pattern of relative densities, and increasing returns help determine how sharply differentiated it is.

## The methods lesson

Two shocks pointing in opposite directions hit the same phenomenon, persistence, and the pair together identifies the model.

- Davis and Weinstein alone would let you conclude that fundamentals are everything. Bleakley and Lin alone would let you conclude that history is everything. Both conclusions would be wrong.
- Compare Week 3. Conley and Kelly showed that persistence established by cross-sectional map correlations often does not survive standard errors that allow for spatial correlation. Persistence established by a shock is much stronger evidence.

When you design your own paper, ask what second design would point the other way.

## How to read a spatial persistence paper

The fifth and last of the weekly checklists.

1. **What kind of shock?** Temporary or permanent. To the factors or to the fundamentals. The answer determines what the result can prove.
2. **What does each theory predict for this shock?** Write it down before looking at the result.
3. **Is persistence identified by the shock,** or only by serial correlation with the past?
4. **How heterogeneous are the fundamentals here?** Japan and Georgia give different answers for a reason.
5. **Sunk costs or increasing returns?** Does the effect decay like depreciating capital, or grow like agglomeration?

Question five is the one Figure V answered tonight.

## What the two papers imply for policy

- **Rebuilding after disaster.** If cities recover on their own, aid should go to people rather than to places. If they can be tipped into a bad equilibrium, the reconstruction of a place is the point.
- **Place-based policy.** Enterprise zones, regional development, relocating a government agency to a struggling city. All of these bet that a push can move an equilibrium. The two papers suggest the push has to be very large, and that whether it works depends on how good the alternative sites are.
- **Declining regions.** Agglomeration cuts both ways. The same forces that keep Augusta on the map keep activity away from places that never got a start.

These are open questions, and we will argue them in class.

## The midterm

- **No class on 12 October**, which is fall break.
- **Oral midterm examinations, Monday 19 October.** Scheduling and the meeting details are in the syllabus.

How to prepare. The five weekly checklists are the exam. Weeks 2 through 6 each ended with one, and between them they cover everything I can reasonably ask.

For every paper on the syllabus so far, be able to state four things.

**Question. Design. Finding. Weakness.**

You may have your notes and the readings open during the exam and you will be asked to explain each answer.

Expect at least one question that asks you to put two papers against each other, such as Allen against Clark or Davis and Weinstein against Bleakley and Lin.

Two weeks to prepare, and every paper since Week 1 is fair game. Start now and come to office hours.